SKILLS

EXCEL

Assignment



Dataset_link: [Corporate_data]

Q1. Customer Classification

The Sales team wants to identify the value of each transaction based on the **Sales Amount, Discount Percentage, Customer Rating, and Order Status.**

Create a new column named **Customer Classification** and classify each record according to the following rules:

Classification	Category
Platinum	Sales ≥ ₹1,00,000, Rating ≥ 4, Discount ≤ 10%, and Order Status = "Completed"
Gold	Sales between ₹50,000 and ₹99,999, Rating ≥ 3, and Order Status = "Completed"
Silver	Sales below ₹50,000 and Order Status = "Completed"
Blacklisted	Any order with Rating ≤ 2 or Order Status = "Cancelled"

Requirements:

1. Insert a new column named Customer Classification.
2. Write a single formula to classify every record.
3. The formula should automatically work for all 500 records.

4. Do not use helper columns.

5. Ensure that every record falls into exactly one classification.

Q2 : Customer Data Standardization

The company is migrating its customer database to a new CRM system. Before migration, the customer information must be standardized to ensure consistency and accuracy.

Perform the following tasks:

1. Standardize all Customer Names by:

- Removing unnecessary leading, trailing, and extra spaces.
- Converting the names into Proper Case (e.g., aMiT shArMa → Amit Sharma)

2. Create a new column named Customer Code using the following format:

Example:

- Customer Name: **Amit Sharma**
- Customer ID: **C001**
- City: **Delhi**
- Customer Code: **Ami001De**

3. Standardize all Email IDs by:

- Removing unwanted spaces.
- Converting all characters to lowercase.

4. If the email address does **not** contain **"@"**, display **"Invalid Email"** instead of the email.

5. Perform all tasks **using Excel formulas only.** Manua editing of records is not **allowed.**

Q3 : Order & Delivery Analysis

The Operations team wants to analyze customer orders to monitor delivery efficiency and product warranty status.

Using the given dataset, perform the following tasks:

1. Create a new column named **Customer Age** and calculate the current age of each customer based on their **Date of Birth.**

2. Create a new column named **Delivery Time (Days)** and calculate the total number of days taken to deliver each order.

3. Create another column named **Delivery Status** using the following conditions:

- **On Time** → Delivery completed within **5 days**
- **Delayed** → Delivery completed after **5 days**
- **Pending** → Delivery Date is blank

Requirements

- Use formulas only.
- Do not edit the original dataset manually
- The solution should work dynamically for all records.
- No Excel error values (**#N/A, #VALUE!, #NUM!**, etc.) should appear in the final output.

Q4 Interactive Product Lookup System

The company wants to build an Interactive Product Look-up System for the inventory team. Instead of searching through the entire product master manually, the user should be able to enter a Product ID and instantly view all the related product information.

Create a new worksheet named Product Lookup Report and design an interactive lookup interface with the following requirements:

1. Create an input cell where the user can enter a Product ID.

2. As soon as a valid Product ID is entered, the following details should be displayed automatically:

- Product Name
- Category
- Supplier
- Cost Price
- Warranty (Months)

3. If the entered Product ID does not exist in the **Product_ Master sheet**, display **"Product Not Found"** instead of an Excel error.

4. The lookup should update dynamically whenever a different Product ID is entered.

5. Do not manually copy or type product details. The report must retrieve all information directly from the Product_ Master worksheet using Excel formulas.

Q5. Sales Performance Summary Report

The management wants a quick overview of business performance without creating Pivot Tables.

Create a new worksheet named Sales Summary Report and calculate the following:

1. Calculate the **total Sales Amoun**t generated from the Electronics category for Completed orders only.

2. Calculate the total Sales Amount generated by Salesperson **"Neha"** in the **North region.**

3. Find the **Number of orders** where the **Discount Percentage is greater than or equal to 15%** and the Payment Mode is UPI.

4. Find the **Number of customers** who have given a **Rating of 5** and whose Order Status is Completed.

5. Calculate the **Average Sales Amount** for orders placed in the **West** region where the **Quantity is greater than 5.**

Requirements

- Create the report in a separate worksheet named Sales Summary Report.
- Use Excel formulas only.
- Do not use Pivot Tables.
- Ensure the report updates automatically whenever the data in the Sales_Data worksheet changes.

Q6. Sales Statistics Report

The management wants to understand the overall sales performance of the company before preparing the monthly business report.

Create a new worksheet named Sales Statistics Report and calculate the following statistics using the Sales column from the Sales_Data worksheet:

1. Calculate the **Average Sales Amount.**

2. Calculate the **Median Sales Amount.**

3. Identify the **Highest** and **Lowest** Sales Amount.

4. Calculate the **Standard Deviation** of the Sales Amount to measure the variation in sales.

5. Determine the **90th Percentile** of the Sales Amount and briefly mention what this value represents.

Requirements

- Create the report in a separate worksheet named Sales Statistics Report.
- Use Excel formulas only.
- Format all monetary values with appropriate currency formatting.
- Ensure the report updates automatically whenever the Sales_Data worksheet changes.

Q7. Product Performance Analysis

The management wants to evaluate the overall contribution of product sales by considering both the Quantity Sold and the Unit Price.

Create a new worksheet named Product Performance Report and perform the following tasks:

1. Calculate the total revenue generated using the Quantity and Unit Price columns without using the existing Sales column.

2. Calculate the weighted average Unit Price, where Quantity Sold is considered as the weight.

3. Verify whether the calculated total revenue matches the total value in the Sales column. If there is a difference, display the difference amount.

4. Calculate the total discounted amount offered across all orders using the Discount (%), Quantity, and Unit Price columns.

5. Display the results in a neatly formatted summary table.

Requirements

- Create the report in a separate worksheet named Product Performance Report.
- Use Excel formulas only.
- Do not use Pivot Tables.
- Format all monetary values as Currency.
- Ensure the report updates automatically whenever the Sales_Data worksheet changes.

Q8. Financial Investment Analysis

The finance department plans to invest a portion of the company's monthly profit into a fixed investment scheme. They want to estimate the future value of these investments and analyze loan repayment options.

Create a new worksheet named Financial Analysis Report and perform the following tasks:

1. Assume the company invests ₹50,000 at the end of every month for 5 years at an annual interest rate of 10%. Calculate the Future Value (FV) of the investment.

2. If the company wants to accumulate ₹50,00,000 after 5 years, calculate the monthly investment required.

3. The company plans to purchase new equipment worth ₹10,00,000 by taking a loan at an annual interest rate of 9% for 5 years. Calculate the monthly EMI.

4. Round the calculated EMI to the nearest whole number and round the Future Value to two decimal places.

5. Display all the calculated values in a well-formatted summary table with appropriate headings and currency formatting.

Requirements

- Create the report in a separate worksheet named Financial Analysis Report.
- Use Excel financial formulas only.
- Display all monetary values in Currency (₹) format.
- Clearly label each calculated value in the report.

Q9. Data Validation & Data Cleaning

Before preparing reports and dashboards, the company wants to ensure that the dataset is accurate, consistent, and ready for analysis.

Create a copy of the Sales_Data worksheet and rename it as Cleaned_Data. Perform the following tasks:

1. Remove Duplicate Records

- Identify and remove duplicate Order IDs from the dataset.

2. Handle Missing Values

- Replace blank values in the following columns:
  - Order Date → Pending
  - Delivery Date → Pending
  - Salesperson → Not Assigned

3. Standardize Text Data

- Remove unnecessary spaces from Customer Name.
- Convert Customer Name, Category, and Region into Proper Case.
- Correct inconsistent category names wherever required.

4. Apply Data Validation

Create validation rules to ensure that:

- Quantity accepts values only between 1 and 20.
- Payment Mode can only be selected from:
  - UPI
  - Cash
  - Card
  - NetBanking

5. Apply Conditional Formatting

Apply Conditional Formatting to highlight all records where the Discount (%) is greater than 20%.

Requirements

- Perform all cleaning operations on the Cleaned_Data worksheet only.
- Do not modify the original Sales_Data worksheet.
- Ensure the cleaned dataset is ready for further analysis, Pivot Tables, and Dashboard creation

Q10. Interactive Sales Dashboard

The management requires an Interactive Sales Dashboard that provides a quick overview of the company's sales performance and allows users to analyze the data dynamically.

Create a new worksheet named Sales Dashboard and design an interactive dashboard with the following components:

1. KPI Cards

Display the following Key Performance Indicators (KPIs):

- Total Sales
- Total Orders
- Average Sales per Order

2. Charts

Create the following Pivot Charts:

- **Column Chart:** Total Sales by Region
- **Pie Chart:** Sales Distribution by Category
- **Line Chart:** Monthly Sales Trend

3. Interactive Filters

Add the following controls to the dashboard:

- **Slicer:** Region
- **Slicer:** Category
- **Timeline:** Order Date

Ensure that all Pivot Charts and KPI values update automatically when a slicer or timeline filter is applied.

4. Dashboard Formatting

- Arrange all KPI cards, charts, slicers, and the timeline neatly on a single worksheet.
- Apply a consistent color theme and appropriate chart titles.
- Hide gridlines to create a professional dashboard appearance.

Requirements

- Create the dashboard in a new worksheet named Sales Dashboard.
- Use Pivot Tables as the data source for all charts.
- Connect all slicers and the timeline to every Pivot Table on the dashboard.
- Ensure the dashboard is fully interactive and updates automatically when filters are changed.
- The final dashboard should be suitable for presentation to senior management.